Blur-Optimum vs. Acuity-Optimum Aperture

Mapped Individually at Each Myopia Level

Same fresh, corrected-optics data as `Fresh_Sweep_Analysis.pdf` (`new_sweeps/blur_sweep.csv`, `acuity_sweep_stage4.csv`) — no new sweep run, replotted per myopia level.

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Chapter 1
What this document shows

`Fresh_Sweep_Analysis.pdf` showed the blur-optimal and acuity-optimal aperture *as a function of* myopia M on one summary chart (its Figure 4). This document instead maps, **individually for each tested myopia level**, exactly where each optimum sits on its own full curve — the blur curve ($r_{EE}(50\%)$ at the retina) and the acuity curve (predicted logMAR), aperture on the x-axis, with a star marking the minimum and a dashed guide line at each optimum's aperture so the gap between the two is visible directly, not just as a single number in a table.

Grid: the same 15 apertures × 8 myopia levels ($M \in \{0, 1, 2, 3, 4, 5, 6, 8\}$D) as `Fresh_Sweep_Analysis.pdf`, same corrected optics (`stage2_fixed/`), same acuity model (`stage4_human_vision`'s recommended predictor). No aperture recommendation is made anywhere below.

1.1 Summary table

M (D)	Blur-optimal (μm)	$r_{EE}(50\%)$ there	Acuity-optimal (μm)	logMAR there	Gap
0	700	7.33*	2000†	−0.344	+185.7%
1	700	7.33*	1250	−0.120	+78.6%
2	750	8.01	750	0.076	0.0%
3	600	10.57	750	0.112	+25.0%
4	500	12.02	600	0.181	+20.0%
5	500	13.29	600	0.240	+20.0%
6	400	14.47	500	0.259	+25.0%
8	400	17.87	400	0.331	0.0%

Table 1.1: *Inside the $r_{EE}(50\%)$ histogram bin-width floor (`Fresh_Sweep_Analysis.pdf`, Ch. 2) — lower-confidence. †Boundary artifact, not a genuine interior minimum — see the note below and Figure 2.1. Gap = (acuity-optimal − blur-optimal)/blur-optimal.

A caveat found while building this document, stated plainly: at $M = 0$D specifically, the acuity (logMAR) curve is still monotonically *decreasing* at the largest tested aperture (2000 μm, Figure 2.1) — it never turns back up within the tested range. The reported "acuity-optimal: 2000 μm" at $M = 0$D is therefore the *edge of the tested aperture range*, not a genuine interior minimum, and the true acuity optimum at $M = 0$D may lie beyond 2000 μm (untested here). This was checked directly, not assumed: every other myopia level ($M \geq 1$D, Figures 2.2–2.8) shows the acuity curve turning back up well inside the tested range, a genuine interior minimum. $M = 1$D's acuity curve turns around at 1250 μm, comfortably inside the 2000 μm ceiling, so only the $M = 0$D row above should be treated as boundary-limited.

Chapter 2
Individual myopia-level maps

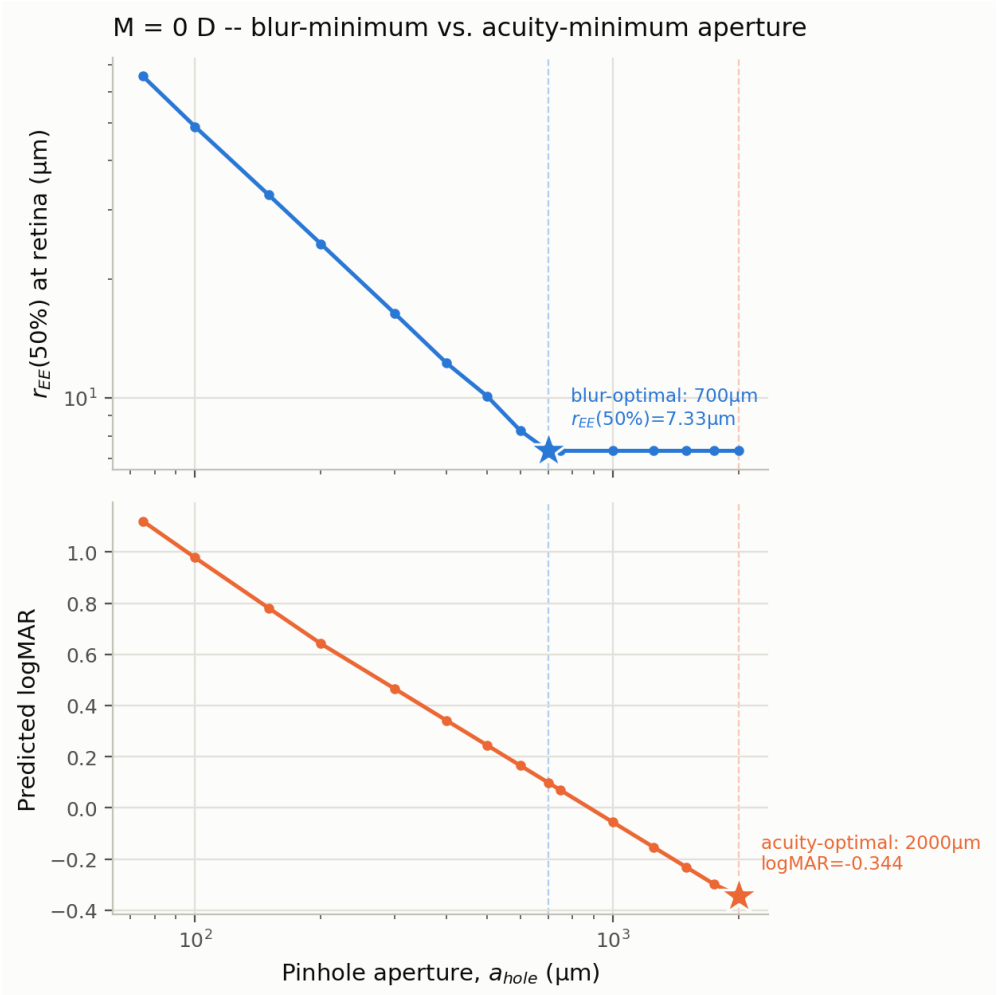


Figure 2.1: $M = 0$D (emmetropic baseline). See the boundary-artifact note above — the acuity curve has not yet turned back up by 2000 μm.

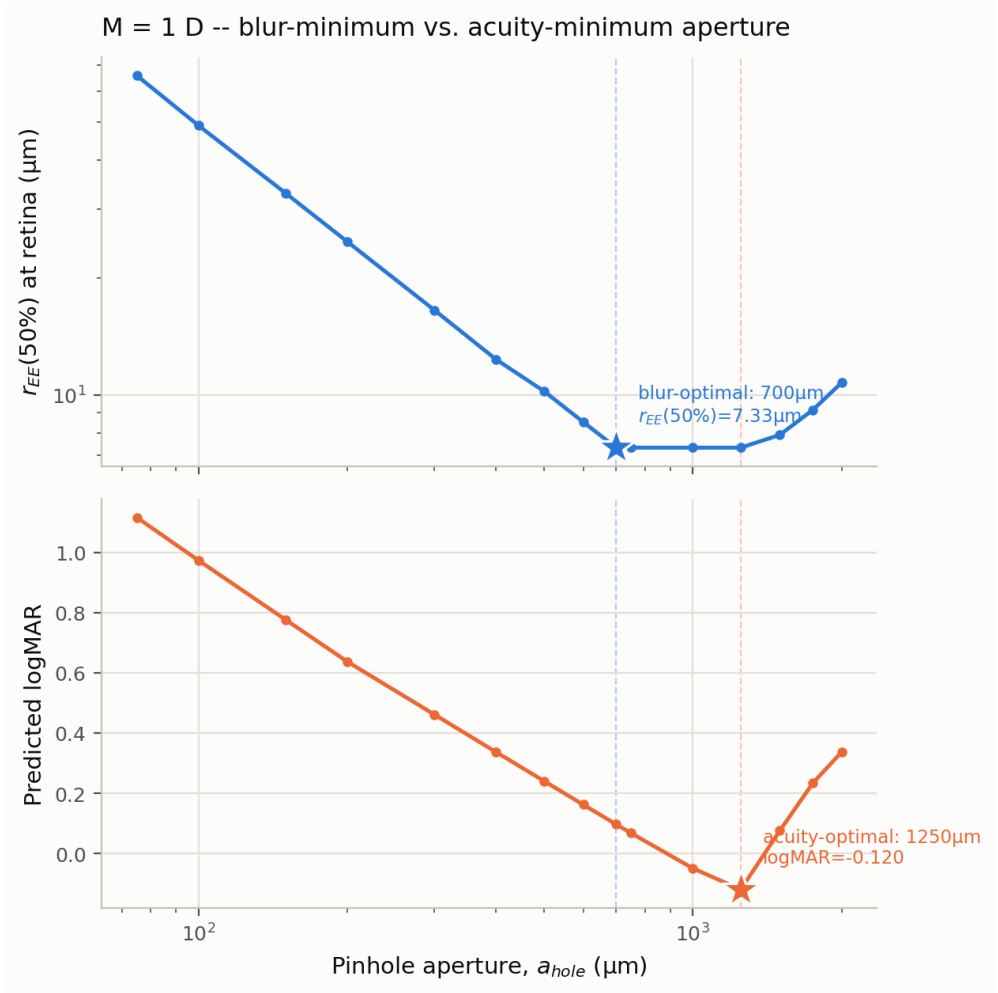


Figure 2.2: $M = 1$D. Both curves show a genuine interior minimum.

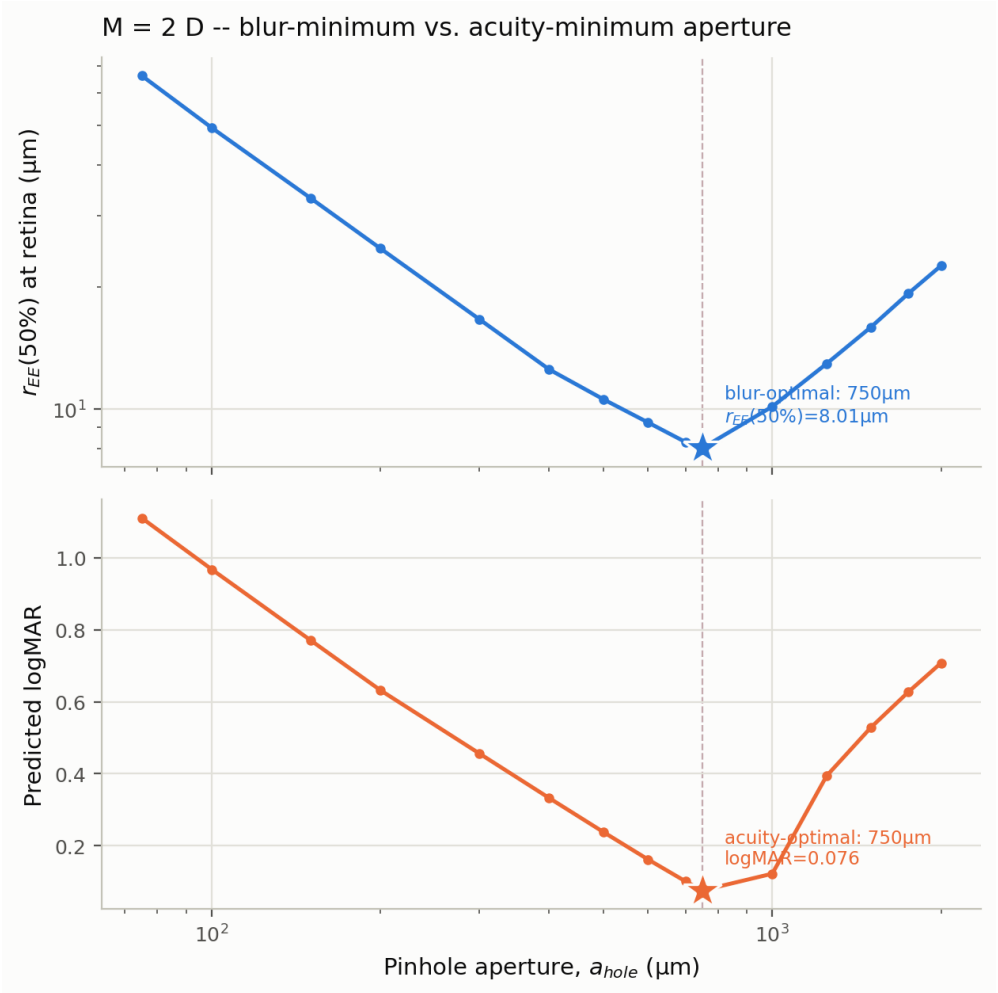


Figure 2.3: $M = 2$D. Blur-optimal and acuity-optimal coincide exactly ($750\,\mu$m).

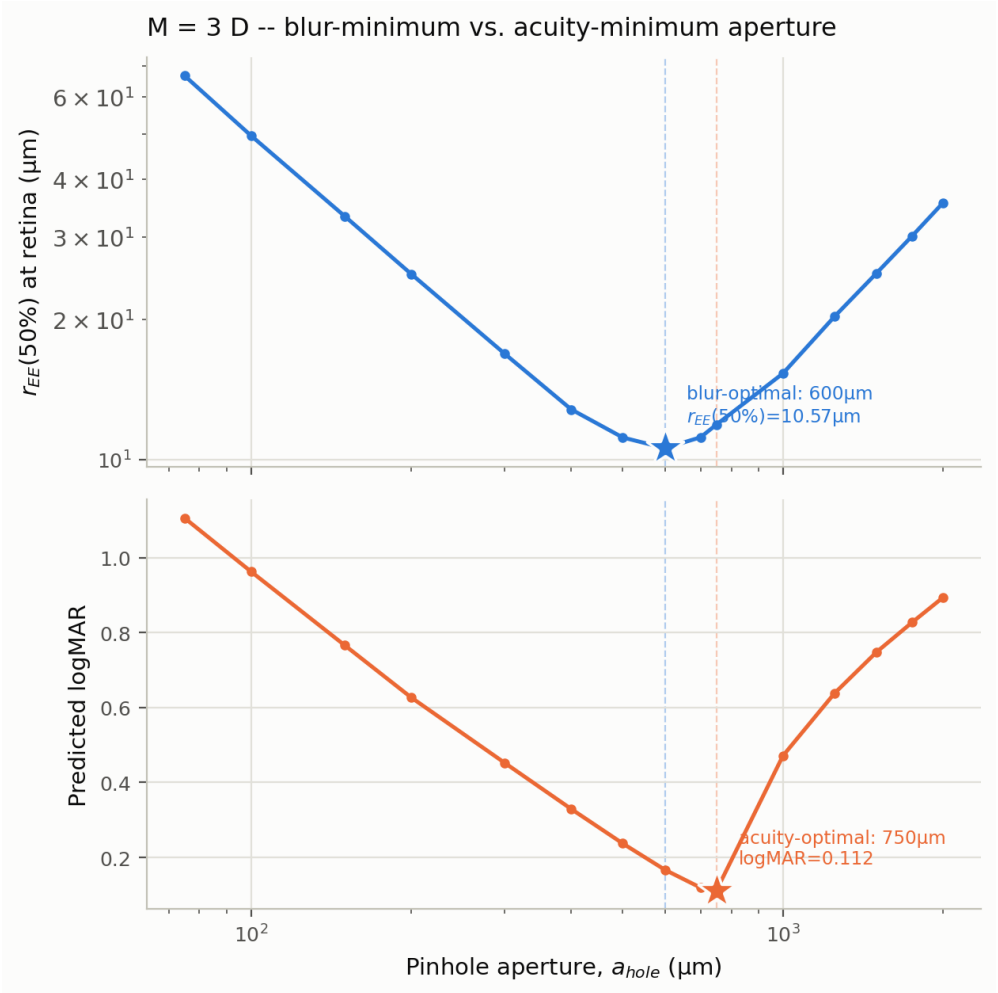


Figure 2.4: $M = 3$D.

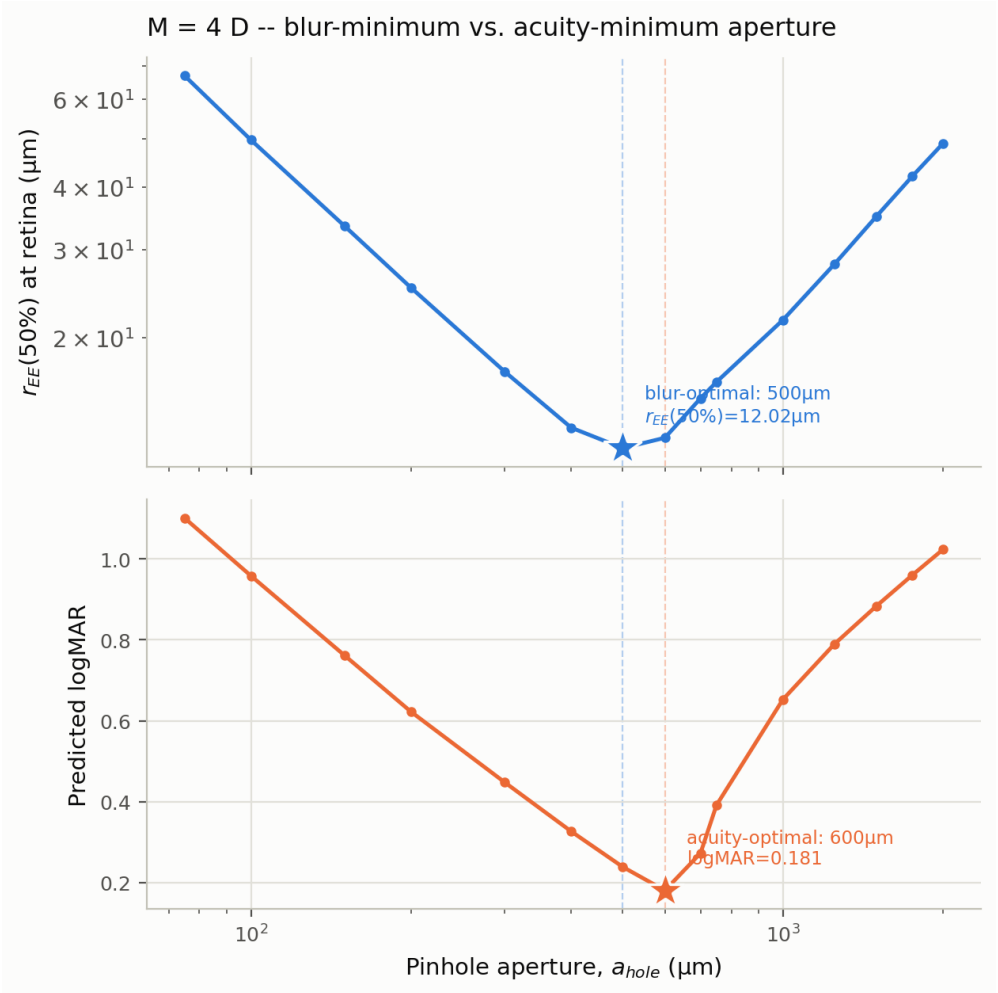


Figure 2.5: $M = 4$D.

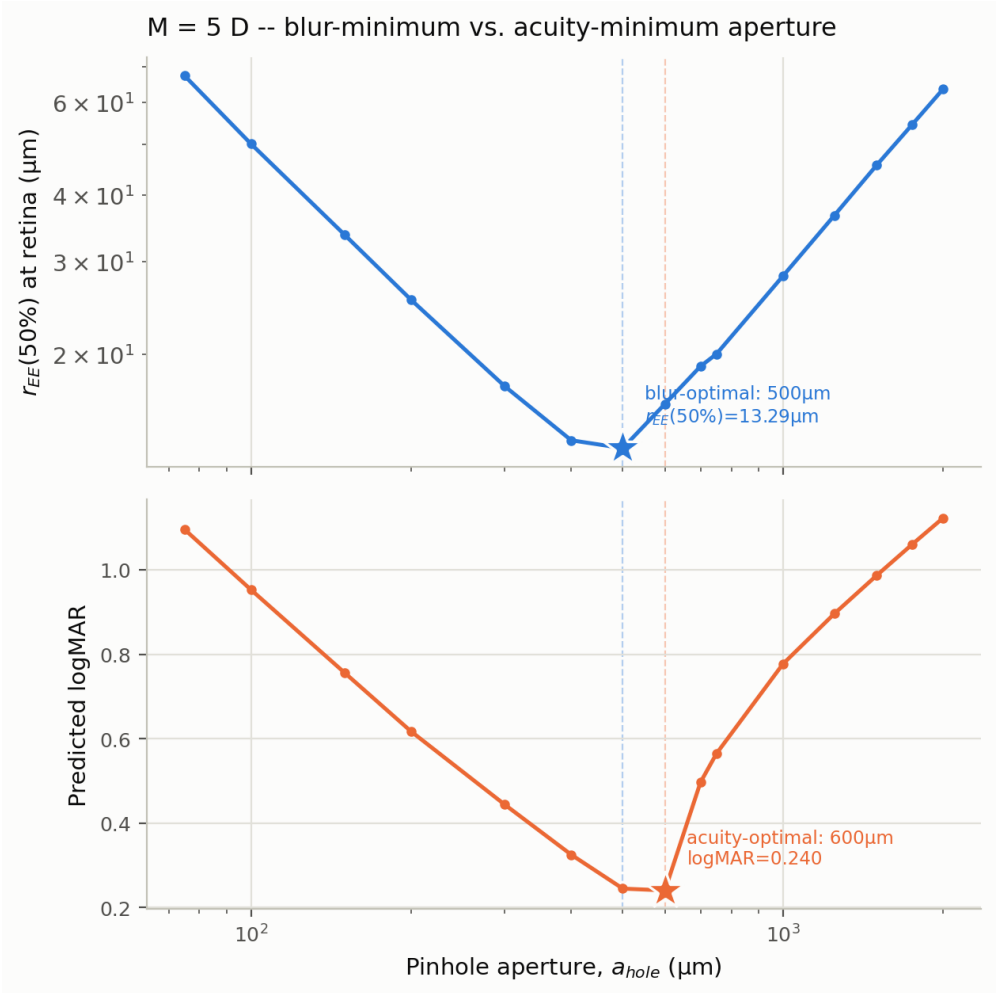


Figure 2.6: $M = 5\text{D}$.

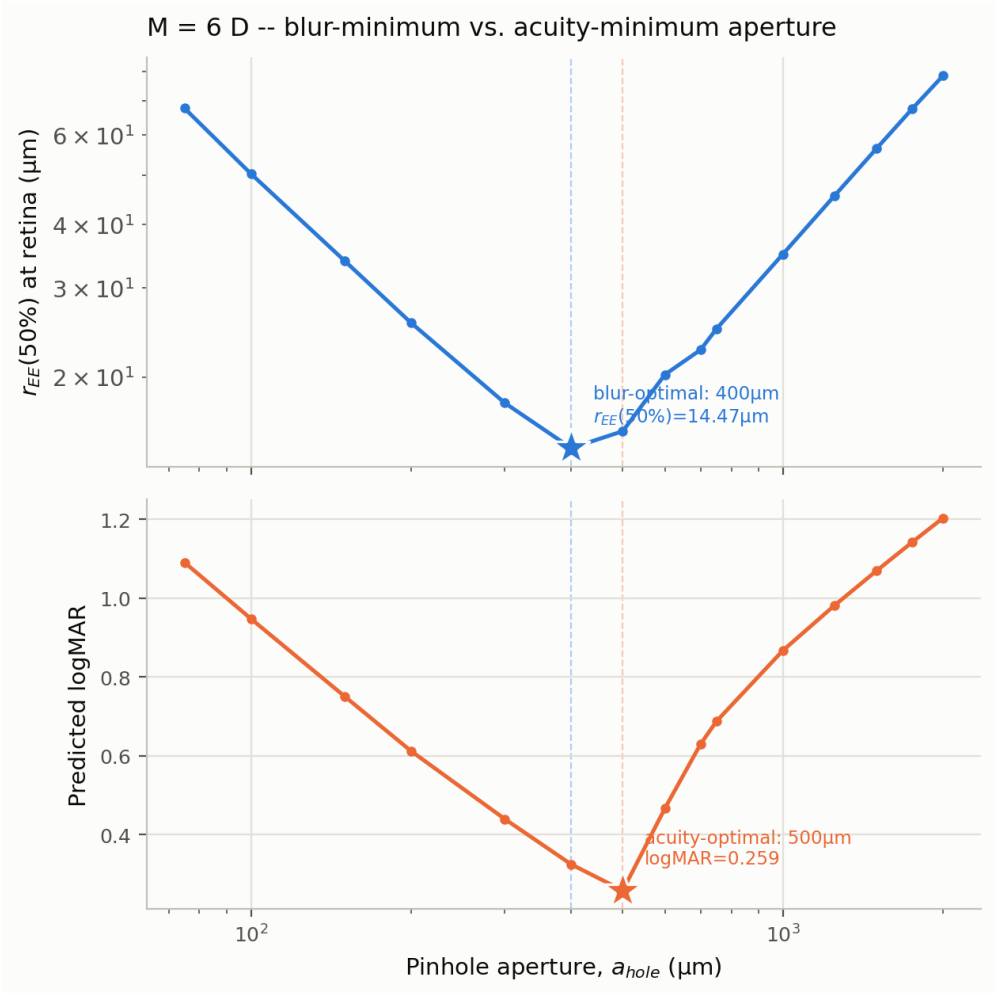


Figure 2.7: $M = 6\text{D}$.

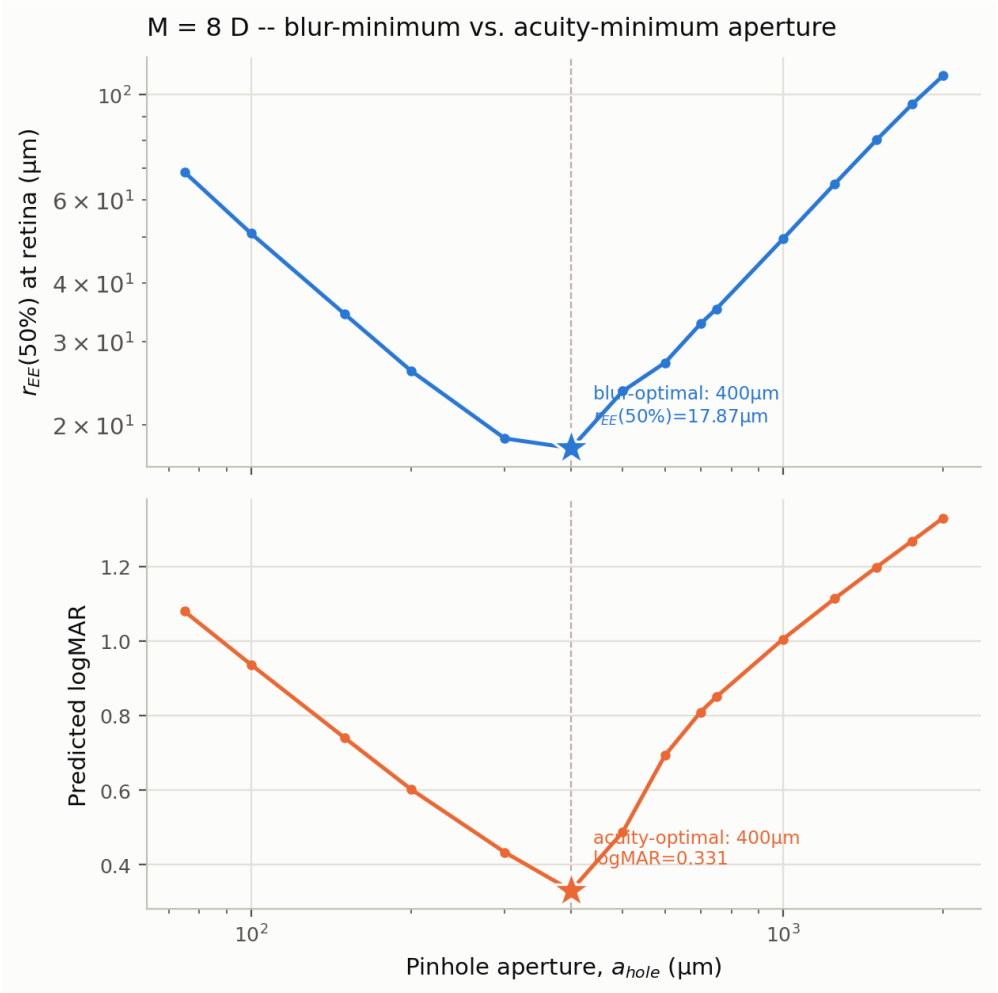


Figure 2.8: $M = 8$D (highest myopia tested). Blur-optimal and acuity-optimal coincide exactly ($400\,\mu$m).

Chapter 3

Reading across the maps

The gap between the two optima shrinks from +185.7% at $M = 0$D (caveated above) to exactly 0% at $M = 8$D, and the two curves visibly converge in shape as M increases (Figures 2.2–2.8): at low myopia the acuity curve's minimum is far broader and shifted well to the right of the blur curve's minimum, while at high myopia the two minima sit directly on top of each other. This is consistent with defocus blur increasingly dominating both metrics at high M (both models are then responding to the same effect), while at low M the acuity model's additional illuminance/contrast term still has room to reward a larger aperture that pure blur minimization does not. No aperture recommendation is made anywhere in this document.